

Chemoinformatic Analysis and Machine Learning Model Development using BCR-ABL Kinase Structurally Diverse inhibitors for Cancer Therapeutics

BCR-ABL is a fusion gene, which is derived from the Philadelphia chromosome abnormality, and is frequently linked to specific forms of leukemia, most notably Chronic Myeloid Leukemia (CML). With an aberrant structure BCR-ABL is a mutant protein that unchecked cells in leukemia. BCR-ABL inhibitor drugs are the exact opposite of curing leukemia by blocking the processing of this spurious BCR-ABL molecule. Such inhibitors will attach the active site of the protein and deregulate its role in the leukemia cells and preventing their growth. In this study, an in-silico approach will be carried out utilizing different machine learning (ML) algorithms implemented in Google Colab. The ChEMBL database provided a data set with 1649 diverse classes of compounds with anti-cancer activity (IC₅₀) targeting human BCR-Abl. Subsequently, SMILES notation for all the compounds was collected and calculated molecular properties to build different ML models. In the present study, PaDEL-Descriptor was applied to compute the data set's fingerprints. 12 classes of molecular fingerprints were used in the current ML study: E-state, MACCS, PubChem, AtomPairs 2D count, AtomPairs 2D, Klekota-Roth, Klekota-Roth count, CDK fingerprinter, extended, graph only, and substructure count and substructure. ML models were built for each of these fingerprints and substructure count fingerprints model was selected as the best classification model based on the accuracy score and other parameters including correlation matrix, and VIP plot. Further, the best ML model was validated using external dataset and based on multivariate and applicability domain analysis. The most active molecules obtained from this model were subjected to molecular docking technique at the active sites of Abl protein. Three molecules were finally selected; F6548-0900, F3216-1016 and F0882-0174 showing better binding affinity with Abl active site residues and also good docking scores. Among these F6548-0900 showed the highest docking score of -10.924 kcal/mol and was selected as the best compound followed by ADME prediction. The present study identified three possible lead compounds for validation using biochemical experiments to develop small molecules for cancer therapy.

Keywords BCR-ABL, ML-QSAR, E-state Fingerprint, MACCS Fingerprint, PubChem Fingerprint, AtomPairs 2D count Fingerprint, AtomPairs 2D Fingerprint, Klekota-Roth Fingerprint, Klekota-Roth count Fingerprint, CDK fingerprinter Fingerprint, Extended Fingerprint, Graph only Fingerprint, Substructure count Fingerprint, Substructure Fingerprint, Molecular Docking, Virtual Screening,

A significant portion of this comprehensive study on BCR-ABL kinase inhibitors emphasizes the role these drugs play in treating chronic myeloid leukemia (CML), a cancer brought on by a translocation of the t(9;22) chromosome that creates the BCR-ABL fusion gene.^[1] As a result, a mutation occurs in these cells that makes the tyrosine kinase constitutively active, which spreads the CML illness. In fact, the introduction section of the research project must clearly state the basic role that targeted BCR-ABL kinase inhibitor therapy plays in the effective treatment of CML. It emphasizes that CML is a dangerous malignant illness with a poor prognosis that can develop into acute myeloid leukemia if untreated or out of control.^[31]

The first section of the article discusses the medication imatinib, a TKI that significantly improved patient outcomes. Conversely, its effects gradually diminish in the following ways. Clinical research indicates that the 5-year average survival rate is 89%, significantly higher than the 31% that existed prior to the development of imatinib. Notwithstanding the imatinib treatment's comparatively high response rate, drawbacks have surfaced. Treatment with imatinib has shown that the medication is less useful in treating CML in its accelerated phase. Additionally, it was unable to totally eradicate the primitive, quiescent leukemia stem cells, or LSCs. Drug resistance and disease relapses, which are frequently chronic, are topics of conversation. This highly contagious disease is frequently associated with inadequate hygiene. Nevertheless, at 60-month follow-ups, undesired resistance rates were found to reach as high as 6% and relapse rates to range from 11% to 14% ^[31]. Scientists and researchers have worked very hard to develop the second generation of TKIs, such as dasatinib, and the third generation TKI, ponatinib, in response to this resistance. Reducing the ability of evolution reactions to develop imatinib resistance and increasing the activity of inhibitors against CML-related kinases is, on the one hand, the primary goal. This also revealed how well these TKIs worked against the BCR-ABL imatinib resistance mutation. In actuality, CML patients have access to a greater range of treatment options, which may lead to improved clinical outcomes. However, it also raises issues related to the treatment of TKIs. Unintended interactions, or side effects related to the heart, have been identified as possible side effects of treatment that may occur during the course of treatment and are caused by activation of the other kinases ^[2]. It creates a need for medications with a multifaceted approach that could be designed to treat the illness holistically while posing the fewest risks to the patient's health. It is easy to understand that the development of BCR-ABL kinase inhibitors requires significant effort from society due to the breadth of these issues. The research idea that this modality should be extremely efficient in addition to being safe is high on the agenda. With this, the therapy enters the medical sphere, and positive results and few side effects are anticipated. This study concretely demonstrates that BCR-ABL kinase inhibitors play the critical role of the main pillar to CML treatment success.^[33] By focusing on the most productive inhibition of activities that occur specifically for each of the mechanisms by which BCR-ABL is active, and which in turn provides for few off-target effects, such an approach is expected to facilitate the treatment. In summary, the introduction mainly fulfilled its purposes in providing the basis for the readers to get ready for the coming treatment options including the applying inhibitors and therapeutic strategy.

Results and Discussion

Selection of Ligands:

The datasets used in this work are included in the ChEMBL Database, together with their Simplified Molecular Input Line Entry System (SMILES) structures and IC₅₀ values in nM. To provide a uniform distribution of the IC₅₀ data, the 1649 compounds' IC₅₀ values were converted to the negative logarithm scale (pIC₅₀). Therefore, $pIC_{50} = -\log_{10}(IC_{50})$ was finally used in the study.

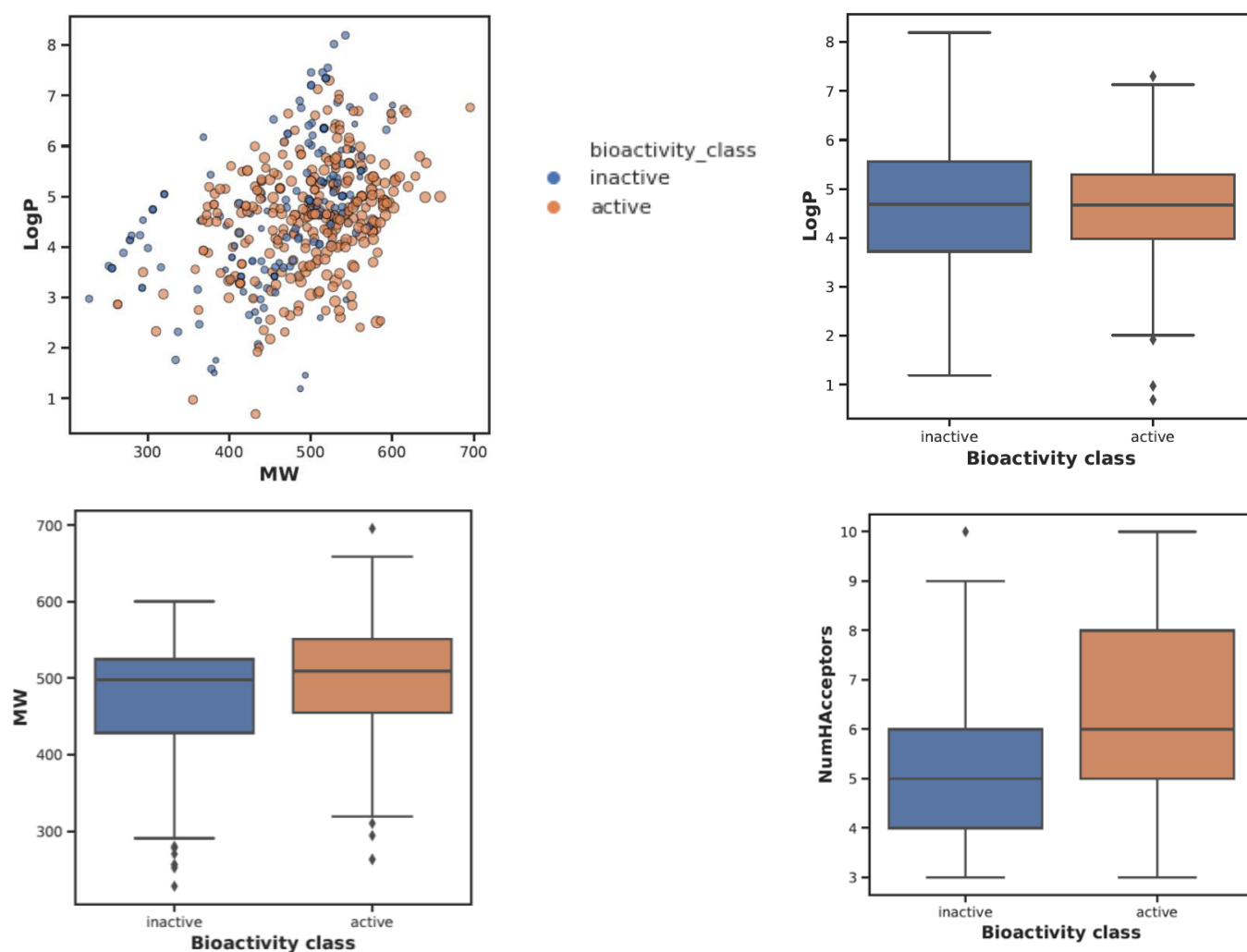
Exploratory data analysis:

Described by the five molecular descriptors from Lipinski rules such as the molecular weight, Log P, the number of hydrogen bond acceptors, and the number of hydrogen bond donors, all small molecules Correlation plots were constructed from the correlation matrix containing different molecular descriptors and bioactivity (pIC₅₀ value) values using matplotlib and Seaborn libraries to graphically demonstrate it. Mann-Whitney U test served to reveal the

significance between two groups, active and inactive. In the case of our data for the active compounds against BCR-ABL showing more significance than inactive compounds the graphic bar plot comparing was adopted (Figure 6 and Table 1)

Molecular feature exploration:

We used twelve molecular features of the 474 BCR-ABL inhibitors we had and the key features were extracted with PaDEL-Descriptor software in order to create QSAR model that could be easily interpreted. The 474 molecules were treated separately for their PubChem fingerprint, Substructure fingerprint, AtomPairs 2D Count fingerprint, AtomPairs 2D fingerprint, fingerprinter fingerprint, extended fingerprint, graph only fingerprint, E-state fingerprint, Klekota–Roth count fingerprint, Klekota–Roth fingerprint, MACCS fingerprint, Substructure count fingerprint. Next, we elected the same features and their proper pIC50 values and SMILES representation. Altogether we used 881 molecular properties from the original PubChem fingerprint property set.(Table 2). Similarly, 307 features were calculated for both Substructure and Substructure count fingerprint, 166 features for MAACS fingerprint, 4860 molecular properties for both Klekota-Roth and Klekota-Roth count fingerprint, 1024 features for Graph-only fingerprint, Fingerprinter fingerprint, and Extended Fingerprint, 79 properties for Estate fingerprint, 780 features for both Atom pairs 2D fingerprint, Atom pairs 2D count fingerprint. The variance threshold as a descriptor selection method was able to point out 36 molecular descriptors as the most significant ones.



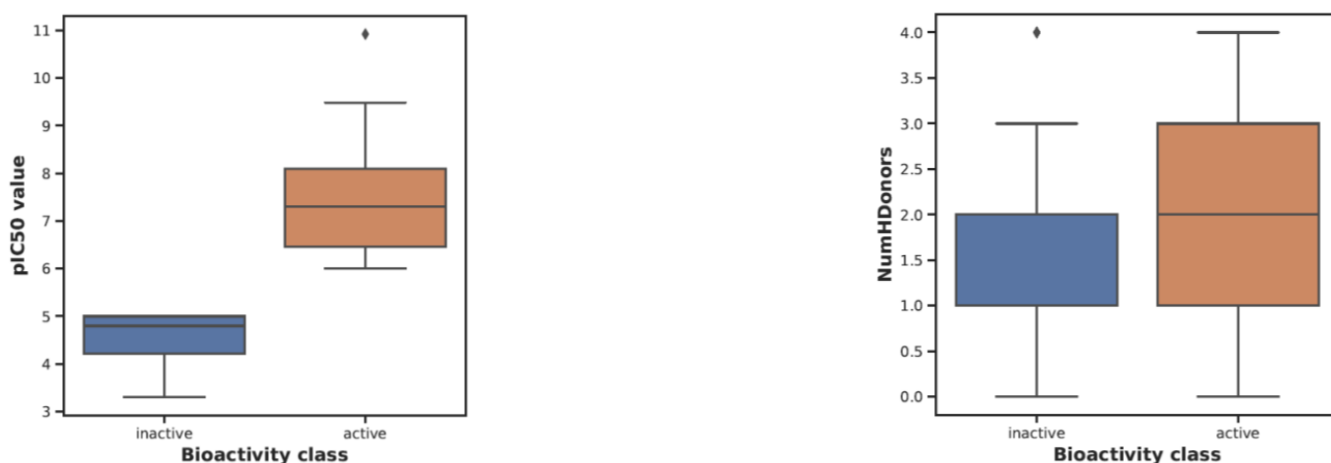


Figure 6: Exploratory data analysis of the BCR-ABL Kinase inhibitor dataset

Table 1: Mann-Whitney U Test results of various properties

Properties	Mann-Whitney U test results	P-value
Molecular weight (MW)	Different distribution	5.37136E-05
LogP	Same distribution	0.450343204
No. of hydrogen bond donor(nHBDon)	Different distribution	2.76E-14
No. of hydrogen bond acceptor(nHBAcc)	Different distribution	5.64E-15
pIC50	Different distribution	5.99E-75

ML-QSAR Model using applicability domain analysis:

Extracting both internal and external sets from the data splits of multiple rounds several times the generalization and interpolation capabilities of the QSAR models were evaluated. The 80/20 ratio was applied to every model in order to split data into two: 80 percent of the training dataset and 20 percent of the test dataset. The QSAR model was next checked using the metrics of Accuracy, MCC, and F1. The first datasets being 80% are used for internal validation and then the second ones 20% are used for external validation. For every model of ML-QSAR the outliers were identified and eradicated through applicability domain analysis. We applied the principal component analysis (PCA) to identify and removed 32 molecules with the best random forest based Substructure count fingerprint prediction model as per (Figure 7). Our optimized

Substructure count fingerprint model made on the random forests algorithm showed of accuracy 1.0, MCC 1.0, F1 1.0 for the training sets and test sets of accuracy 0.9714, MCC 0.9406, F1 0.9717 respectively. Substructure count fingerprint model is the best model as per the values of training sets(80% of internal set) accuracy , MCC, F1.

Interpreting the machine learning-QSAR model:

We utilized VIP plots to depict the variance relevance of each machine learning QSAR model's top 10 molecular characteristics. Substructure count prediction model, we extracted SubFPC172, SubFPC100, SubFPC300, SubFPC302, SubFPC301, SubFPC275, SubFPC295, SubFPC274, SubFPC181, SubFPC184. As a consequence of the investigation, we created correlation matrices for the first 10 available chemical characteristics for each machine learning-QSAR prediction model using VIP plots. In addition, it was discovered that a link exists between the ten highest characteristics extracted from each machine learning -QSAR prediction model. All major characteristics retrieved from the VIP plot analysis played important roles in BCR-ABL inhibition.

As shown in Figure 9(a) and Table 3, the top ranking feature is Arylfluoride (SubFPC172), Second top feature is Secondary amide (SubFPC100), Third and Fourth top features are 1,3-Tautomerizable (SubFPC300) and Rotatable bond (SubFPC302), Fifth and Sixth top features are 1,5-Tautomerizable (SubFPC301) and Heterocyclic (SubFPC275), Seventh and Eighth top features are C ONS bond (SubFPC295) and Aromatic (SubFPC274), Ninth and Tenth top features are Hetero N nonbasic (SubFPC181) and Heteroaromatic(SubFPC184).

Table 2: Performance summary of QSAR Model for predicting BCR-ABL Kinase inhibitors

Name	FP No	Descriptors	Final Model					
			Outliers	Training No.	Test No.	Accuracy	MCC	FI
Substructure Fingerprint	307	18	63	345	66	Train: 0.9884; Test: 0.9394	Train: 0.9750; Test: 0.8746	Train: 0.9884; Test: 0.9394
Substructure count Fingerprint	307	36	32	372	70	Train: 1.0; Test: 0.9714	Train: 1.0; Test: 0.9406	Train: 1.0; Test: 0.9717
Pubchem Fingerprint	881	200	14	374	86	Train: 0.9893; Test: 0.9651	Train: 0.9770; Test: 0.9289	Train: 0.9893; Test: 0.9651
MACCS Fingerprint	166	79	30	369	75	Train: 0.9837; Test: 0.9333	Train: 0.9644; Test: 0.8589	Train: 0.9837; Test: 0.9336
Klekota Roth Count Fingerprint	4860	161	31	377	66	Train: 0.9973; Test: 0.9697	Train: 0.9944; Test: 0.9261	Train: 0.9973; Test: 0.9697
Klekota Roth Fingerprint	4860	114	41	370	63	Train: 0.9973; Test: 0.9524	Train: 0.9942; Test: 0.8914	Train: 0.9973; Test: 0.9535
Graph only Fingerprint	1024	395	22	375	77	Train: 0.9893; Test: 1.0	Train: 0.9776; Test: 1.0	Train: 0.9893; Test: 1.0
Fingerprinter Fingerprint	1024	815	15	379	80	Train: 0.9947; Test: 0.925	Train: 0.9887; Test: 0.8508	Train: 0.9947; Test: 0.9243
Extended Fingerprint	1024	820	22	379	73	Train: 0.9921; Test: 0.9589	Train: 0.9832; Test: 0.9135	Train: 0.9921; Test: 0.9583
Estate Fingerprint	79	16	45	353	76	Train: 0.9943; Test: 0.9868	Train: 0.9876; Test: 0.9697	Train: 0.9943; Test: 0.9868
AtomPairs 2D count Fingerprint	780	163	22	379	73	Train: 0.9868; Test: 0.9452	Train: 0.9716; Test: 0.8871	Train: 0.9868; Test: 0.9455
AtomPairs 2D Fingerprint	780	90	27	371	76	Train: 0.9919; Test: 0.9605	Train: 0.9829; Test: 0.9057	Train: 0.9919; Test: 0.9603

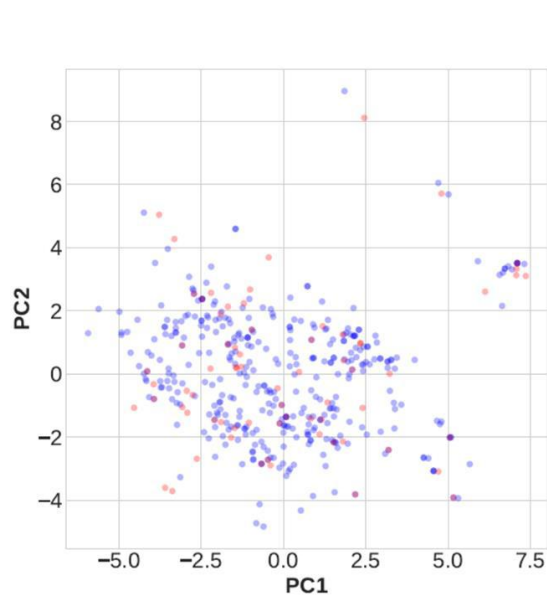


Fig 7(a): 2D PCA of the Substructure count fingerprint prediction model

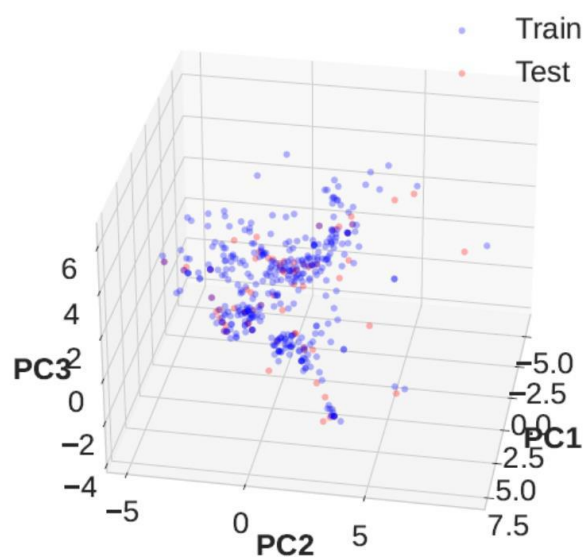


Fig 7(b): 3D PCA of Substructure count fingerprint prediction model

Figure 7: Applicability domain analysis by PCA Plot for the generated machine learning-QSAR model

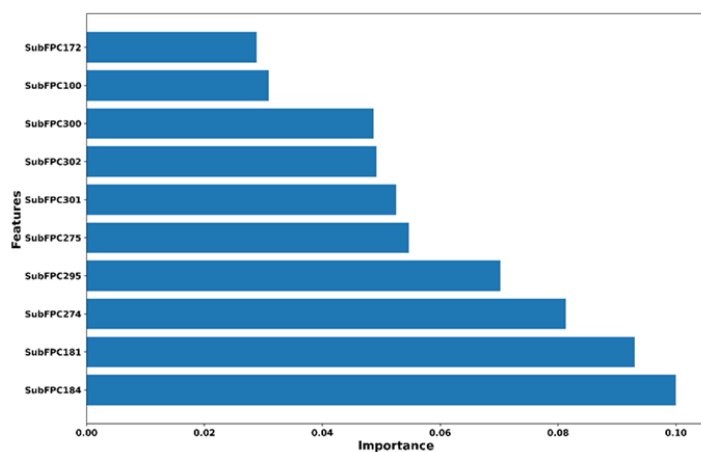


FIG 9(a): VIP Plot analysis

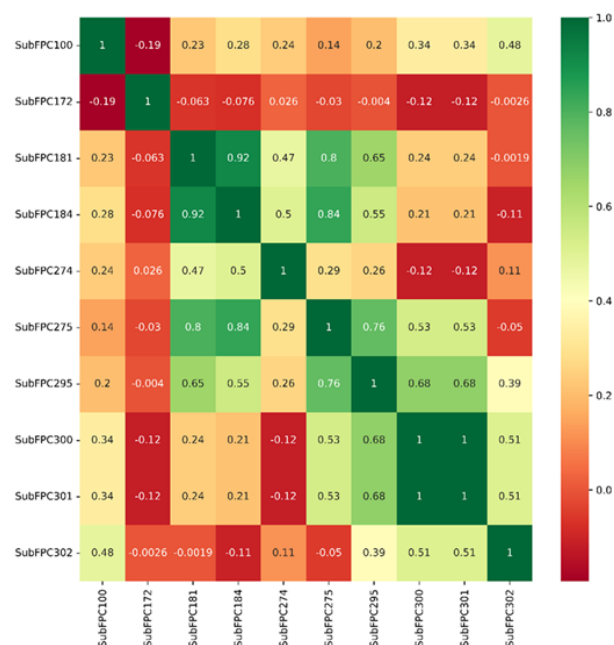


FIG 9(b): Correlation matrix analysis

Figure 9: VIP Plot analysis and Correlation matrix analysis of ten optimal features of the Substructure count prediction model

ROC Curve analysis:

Sensitivity and specificity (false positive rate) together with cut-off values generate the ROC curve which plots the coordinate points with $1 - \text{specificity}$ (false positive rate) as for x-axis and sensitivity as y-axis. Accuracy in a test's diagnosis is usually calculated for the AUC. The better the testing accuracy, the closer is the ROC graph to the top left corner of the graph showing that Sensitivity=1 and the specificity of that test=1 (false positive rate)=0. Hence, AUC has a value of 1.0 that depicts the perfect ROC curve. Therefore, we choose the sub count fingerprint model as an ideal model instead of the other 12 models because the internal or training data set of this model obtained AUC = 1.0 for all metrics, respectively accuracy, MCC and F1.(Figure 10)

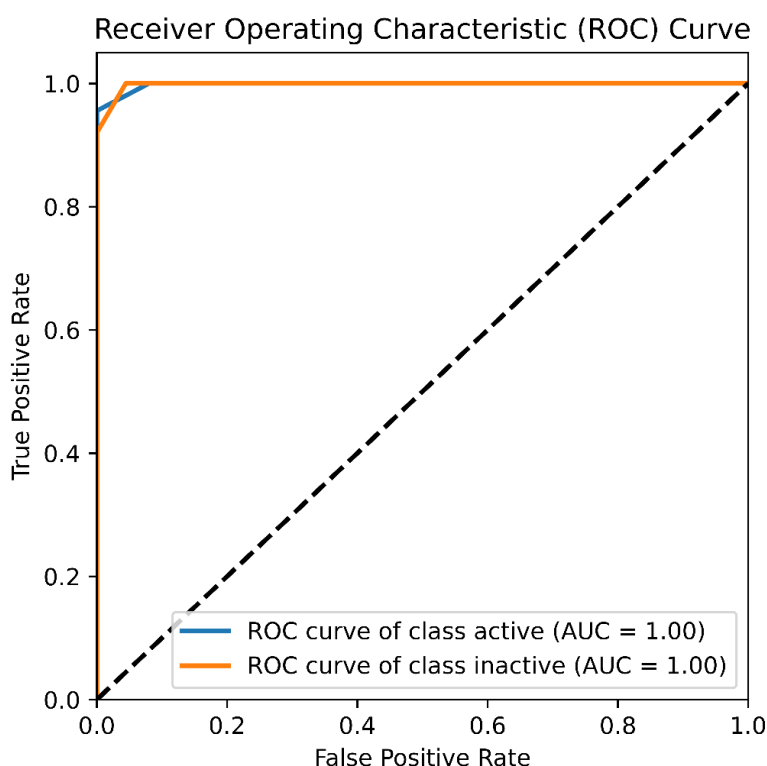


Figure 10: ROC curve graph of Substructure count fingerprint prediction model

Virtual Screening for Identifying best Abl inhibitors:

Using the developed machine learning-QSAR-based web application, we screened an LC Anticancer focused library on the ChemDiv website. 836 molecules were found after a total of 9156 molecules were first screened using the Chembioserver 2.0 web platform based on three drug-likeness rules: Veber rules, Lipinski's rule of five, and Ghose filters. Additionally, as a part of this approach, we also run the twelve fingerprint-based prediction models to predict each molecule individually and evaluate its activity against BCR-ABL. Finally, we were left only with 511 active molecules after taking out all the inactive ones applying the best fingerprint model, these we put through molecular docking analysis.

Table 3: List of the top 10 Substructure count molecular feature interpretation

Fingerprinters	Description
SubFPC172	Arylfluoride
SubFPC100	Secondary amide
SubFPC300	1,3-Tautomerizable
SubFPC302	Rotatable bond
SubFPC301	1,5-Tautomerizable
SubFPC275	Heterocyclic
SubFPC295	C ONS bond
SubFPC274	Aromatic
SubFPC181	Hetero N nonbasic
SubFPC184	Heteroaromatic

Molecular Docking and analysis based on machine learning -QSAR models:

For the ligand-based virtual screening, the LC-Anticancer library was utilized having only 511 active molecules in it. These molecules were then screened using three levels of high-throughput virtual screen (HTVS) (> 8 kcal/mol), standard precision (SP) (> 8 kcal/mol), as well as extra precision (XP). The compound's binding mechanisms and assess how structural changes affect the compound's inhibitory activity against imatinib, docking analyses were carried out. (Figure 11a). In order to do It was possible to obtain the Human Abl kinase domain in complex with imatinib (PDB ID: 2HYY). Among 80 hits, In XP mode the docking scores ranged -8.500 to – 10.580 kcal/mol and highest value -10.924. Interestingly, compound F6548-0900 had about the same binding score, -10.924 kcal/mol, against BCR-ABL. Hydrophobic interactions surrounded by GLU316 and MET318 in BCR-ABL were favourable (Table 11b)

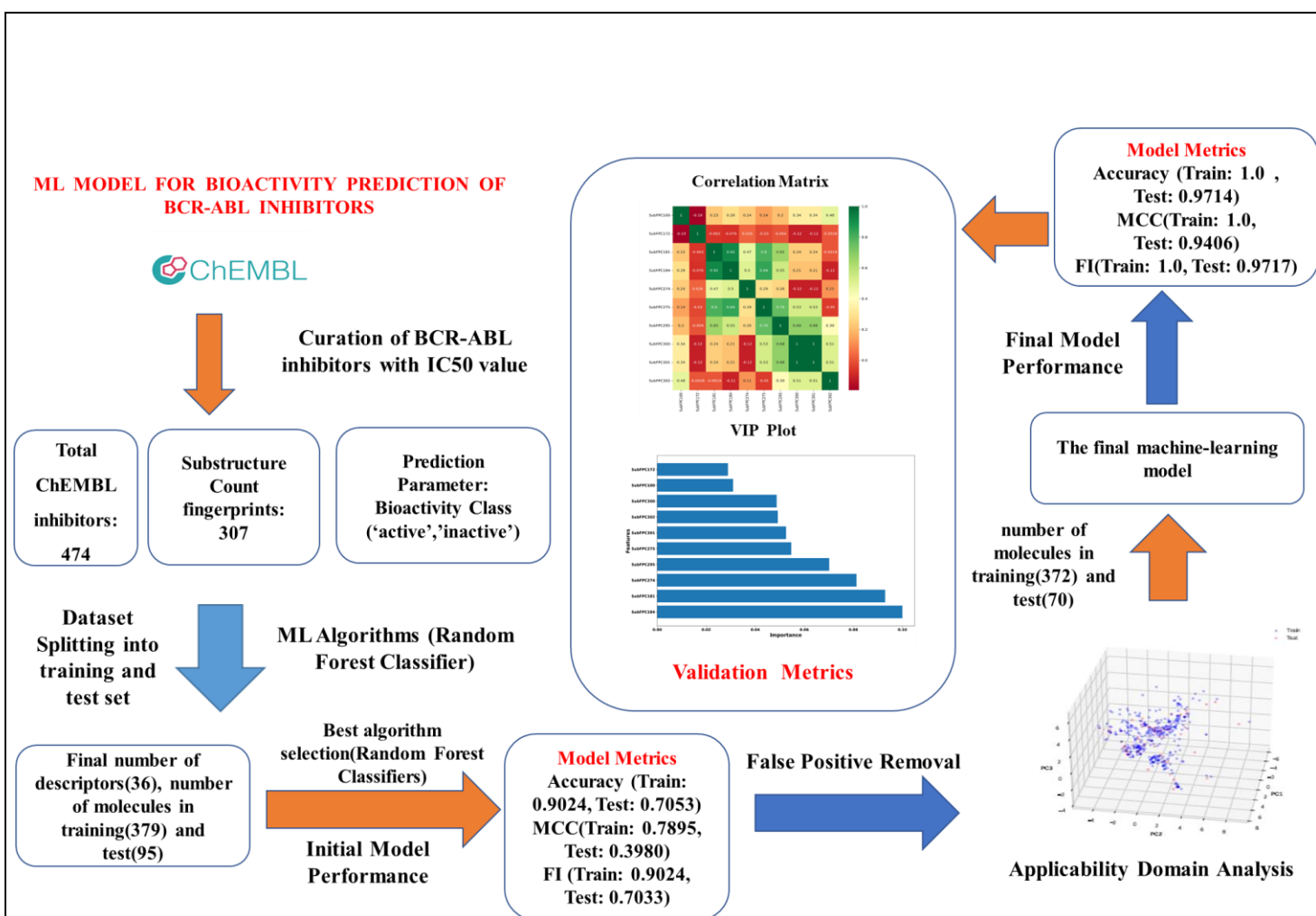


Figure 8: Workflow for the generation of machine learning-QSAR Substructure count model

Determination of ADME/T:

With the help of Schrodinger's Qikprop v.2.3 software, the best molecules from top 10 were examined. From these, specific characteristics which are highlighted here, are critical in determining which substances would be suitable as therapeutic agents. As per (Table 4), These characteristics are:

- Molecular weight of the molecule (mol_MW)
- Octanol/water partition coefficient (log po/w)
- Dipole: Computed dipol moment of the molecule
- SASA: Total solvent accessible surface area

- FOSA: Hydrophobic component of SASA
- FISA: Hydrophilic component of SASA
- Volume: Total solvent accessible volume
- Number of Hydrogen Bond donors (donorHB)
- Number of Hydrogen Bond acceptors (accptHB)
- Percent human oral absorption
- PSA: Van der Waals surface area of polar nitrogen and oxygen atoms
- Number of violations of Lipinski's rule of five

According to the Lipinski rule of five, all of the structures illustrated drug-like features with noteworthy values for the properties investigated. In order to be classified as a drug-like molecule, a compound needs to meet at least one of the following requirements: its molecular weight should be not more than 500 Da, its octanol–water partition coefficient should not more than 5, its hydrogen bond donor should not more than 5 and its hydrogen bond acceptor should not more than 10. And for Percent human oral absorption it should not higher than 80% but as per Table 4 we observed that five compounds got 100% value. This table shows a number of physicochemical properties computed by Qikprop. SASA, FOSA, FISA and Volume were also predicted from the surface area components.

CONCLUSION

The emergence of new BCR-ABL kinase inhibitors for the treatment of cancer shows the potential of machine learning and chemoinformatic analysis can be useful in identifying novel molecules that are structurally diverse yet potent against the BCR-ABL protein, which is a key driver of ALL and CML that is chromosome-positive. Our study aimed to develop a

multiple-feature machine learning -QSAR model using the ligand-based drug design approach to determine the structural features of the BCR-ABL kinase inhibitors that underlie their bioactivity mechanism. We performed twelve molecular feature-based machine learning-QSAR models using twelve differing fingerprints. The final Substructure count model showed the Accuracy values 1.0 and 0.9714, MCC values 1.0 and 0.9406 and FI values of 1.0 and 0.9717 for the molecules train and test set that contained 372 and 70 molecules, respectively ML algorithm is random forest. We further screened a LC Anticancer focused library of 9156 molecules, F6548-0900, which is the highest binding affinity inhibitor identified was revealed to be the most efficient towards Abl kinase in the present study. The purpose of this was to find this out using a complete structural analysis contributing essential molecular features selected from machine learning-QSAR models, molecular docking and analysis of interactions. Machine learning algorithms show the ability of detecting new BCR-ABL inhibitors from a large database of small molecules with huge structural variation. Illustrates the rationale of the strategy for identification of novel treatment directions in cancer therapy when machine learning algorithms are applied for activity prediction toward the BCR-ABL protein.

METHODS AND MATERIALS

Data Collection:

The chemical library of possible BCR-ABL inhibitors was selected from ChEMBL database (ChEMBL2096618) and consisted of 844 compounds and 1649 bioactivity data points targeting human Bcr/Abl fusion protein. The activity was measured using various methods, mainly expressed as percentage inhibition (%), IC₅₀ (nM), GI₅₀, EC₅₀, PI, AC₅₀, Ratio, FC, K_d. The results of BCR-ABL inhibition were calculated as half-maximal inhibitory concentrations (IC₅₀) in the mode of micromolar units [μ M] that shows amount of a drug required to hit a target or a pathway. This shift enabled the values of the drugs to be represented in a logarithmic scale ($\text{pIC}_{50} = -\log \text{IC}_{50} \times 10^{-6}$) so that the activity values could be made linear and approximate normal distribution. Based on the QSAR study, pIC_{50} was selected as a sensitive variable to eliminate values variability.

We classified the data used for machine learning model development as active, inactive or intermediate using the following criteria —Active : IC₅₀ value < 1000 nM, Intermediate : $1000 \leq \text{IC}_{50} \text{ value} < 10,000 \text{ nM}$, Inactive : IC₅₀ value > 10,000 nM. The values less than 1000 nM for the most outstanding inhibitors served as the threshold values as it was the better models that were created with this threshold value. In addition, as inverse type II drugs, levels of activity as inhibitors are not used for evaluation if IC₅₀ value is $10^{[9]}$. As a result, the indicator was rated as intermediate and inactive when value was increasing. Smaller the IC₅₀ value, lesser concentration is needed for inhibition. Our research targeted the intermediates during the process. Classification of dataset performed by both the active and inactive chemicals in our models. In the next step, the BCR-Abl dataset was cleaned by removing the similar species with (i) identical SMILES notation and (ii) missing IC₅₀ values ^[9].

Molecular descriptors calculation:

The 3D structures of compounds must be translated into a computer-processable form so that machines can comprehend and the computer can process them. The algorithm calculates reaction specific parameters like molecular desirability. They have been associated to chemicals to convey their bio activity before the research progress. In the second step, a QSAR model is generated using these descriptors. Descriptors values are specific to molecule properties and are used to facilitate different tasks, e.g., similarity search and drug design.^[37] While considering the descriptors of deep learning model, it is essential considering the realistic visual input that requires accuracy and care. Physical and chemical characteristics are the common terms we use to describe how molecules behave with respect to solubility, lipophilicity, and molecular weight. The chemical composition could have a prominent impact upon the medication production process too. On the other hand, disregarding molecular geometry and looking at the bond connections give the molecular descriptors a more significant and global concept for with molecules are identified. Prior to taking a call on getting a strategy that will fit our purpose we must first thoroughly examine the pluses and minuses of all the strategies.^[20]

PaDEL Descriptors:

PaDEL is a popular molecular description and computational chemistry software tool that is a great option for high-throughput QSAR modeling. Emphasis of PaDEL's on producing a variety of molecular descriptors efficiently, makes it especially useful for experiments. Its quick computation of descriptors for big chemical databases allows for quick screening and analysis, and its intuitive interface makes it accessible to researchers with different levels of experience. Our knowledge of drug design and discovery can be advanced by effectively examining the complex relationships between molecular structures and activities with PaDEL.

The fingerprints of the data set were computed using the PaDEL-Descriptor software. This would help with the structural information analyses and are used to characterize the different properties of compounds. The molecular descriptors are generally very important for QSAR studies. Twelve molecular fingerprints – MACCS, PubChem, substructure count, substructure, AtomPairs 2D Count, AtomPairs 2D, CDK fingerprint, extended, graph only, E-state, Klekota–Roth count and Klekota–Roth respectively^[9].

Lipinski Descriptors:

Lipinski has come up with some of his own filters that characterize the drug-likeness compound of each particular molecule. The study focus on the pharmacokinetic properties of a drug molecule known as the pharmacokinetics, thus enables researchers to know whether an drugs taken orally is a competent compound among other molecules. The elements that are listed below are included in this case. If you incorporate two or more units that deviate from standard requirements in this case, you are not supposed to consume this molecule. [15]. Lipinski's rules could be seen by performing algebraic cardinalities of the molecule's mathematical system in the small molecules. The drugs will have (i) molecular weights ≤ 500 Daltons, (ii) ≤ 10 rotatable bonds, (iii) ≤ 10 hydrogen bond acceptors, and (iv) ≤ 5 hydrogen bond donors.

Veber's Rules:

Veber's rules is compound classification system that comprises of a set of descriptor properties, which assessment of their components' qualities for drugs making. It is the logical continuation of Lipinski's principles. It adds two additional descriptor: (a) the polar surface area, (b) the number of the rotatable bonds.

The amount of relative bonds is an indicator of the flexibility of a molecule, and a high amount of rotatable bonds, in turn, can really decrease the oral bioavailability and the absorption. Rule has a limit of 10 rotatable bonds in single molecule for its molecule to be accepted as drug like structure.^[3] Polar area surface is the surface area that is occupied by a polar molecule. The high polar surface area can become an obstacle to these processes, if not it impairs them completely. By getting this information, a compound should have a polar surface area that is not more than 140 Å². It should be drug-like.

Ghose Filters:

The Ghose filters are another set of an molecular descriptor, which also belongs to the group of properties that are used for the analysis and evaluation of compounds.^[34] It is also acting as an appendix to Lipinski's rule same like Veber's rules. Two additional descriptors: It is also acting as an appendix to Lipinski's rule and Veber's rules. Two additional descriptors:

- Total Atoms ≥ 20 AND ≤ 70
- Molar Refractivity ≥ 40 AND ≤ 130

Data pre-processing:

Before constructing the dataset comparison unit variance and mean centering scaling of descriptors were performed before building the classification model. If the correlation coefficient and pairwise inter-correlation coefficients which were found of the threshold values 0.7m and 0.95m were removed, then the descriptive variables were retained from the model. To realize that process, the smaller subsets were produced by definite descriptors such as AtomPair 2D Count, AtomPair 2D, CDK Fingerprinter, CDK Extended, CDK Graph Only, ES-state, Klekota-Roth Count, Klekota-Roth, MACCS, PubChem, Substructure Count, and Substructure with 120, 154, 31, 951, 934, 599, 405, 452, 92, 196, 64, and 66.^[9]

Data splitting:

It should be added that same constant (or even nearly constant) values for given fingerprint descriptor sets were considered to support the model construction issue to decrease its complexity and avoid the bias. The absolute values for all the fingerprint descriptor thresholds were calculated by the 0.1 standard deviation (SD) cut-off criterion. At this point, the sequences with SD of 0.1 and above were selected for sequencing. The aim was preventing the risk of overfitting in imbalanced data and applying the random subsets of active compounds from the introduction set through the under-sampling technique. In order to eliminate any chance of receiving a biased prediction for the model, the data was additionally split into two sets: for the second dataset, The training data set contained the other 80% and the test data set made up of 20%, while of the information to be processed.^[18]

WEKA 3.8.5:

Weka, a software package that is cited as the most popular toolbox used in ML tasks, has been used in the problem of predicting cancer, and also in the problem of deciding about patient therapy, therefore its role is clearly demonstrated. We have the ability to construct predictive models for regression and classification tasks as well as to clustering. Weka was used in prediction of binding affinities and activities of neither came at the expense of accuracy. It allows to form a gather of the various molecules according to their chemical and physical properties.^[7]

Multivariate analysis of model:

Besides the input factor of a QSAR model and the descriptor of a compound, the other factor that may affect the quality of prediction is the level of predictability. Because RF gives the best result in the real world and easily understandable under the raw forms is the main argument of why we choose it. The classifier type Random Forest sometimes trains random small sections of training data samples and variables and builds several decision trees. At the top of the hierarchical RF classification process is the division of input data at the root node by using good features via a similar process. Subsequently, the voluminous amount of trees pollution data from every tree is considered to identify the classification in the form of the impersonal average result among all trees. A RF classifier was constructed using the R Package Random Forest in the languages R. Two RF model parameters need to be adjusted in order to predict BCR-ABL Kinase inhibitor activity accurately: the number of randomly selected features of trees not to forget about the number of trees also used for the RF classifier.^[20]

Applicability domain analysis:

The applicability domain of the present BCR-Abl QSAR model is determined using the principal component analysis (PCA) bounding box approach. This key theoretical approach by applying PCA analysis of scores plots and to compare the chemical space of chemicals from the external dataset to those from the training set respectively. Sklearn's PCA function was utilized for the present work. Python deconstruction module for scikit-learn machine learning library was also implemented.

Receiver Operating Characteristic curve (ROC):

A receiver operating characteristic (ROC) curve is a graph that clearly shows how well the classification model works across all decision-thresholds in a threshold-independent manner. The AUC curve is an important model evaluation tool since it demonstrates the graphical trade-offs between specificity and sensitivity [11]. AUC is the ability of the classifier which is tested by comparing it with a perfect classifier and a random classifier whose values are 1 and 0.5, respectively. It may be designed to look at how good machine learning model can be at forecasting the resistance of Abl clinical mutations to kinase inhibitors in targeted therapies.

Molecular Docking:

Protein Preparation:

BCR Abl kinase domain in complex with imatinib (PDB ID: 2HYY) structure is taken from the Protein Data Bank (PDB) as seen in Fig. 3. The protein structure was prepared using the

Protein Preparation Wizard (PPW) module of the Schrödinger software package. PPW is a comprehensive process that includes every node from beginning to end; extracting and refining, reviewing and editing, and protecting and reducing is steps in the process. To start off, the proteins was converted into actual amino acids by adding hydrogen atoms, adding water molecules at the active sides and fixing the possible unknown side chains/ residues in it.

Receptor Grid Generation:

Binding pocket description is essential for the molecular docking study. The Receptor grid is from the Schrödinger suite and the default mode is applied. The algorithm took into account the center of mass of centroid and grid generation of the bound compound in the human Abl

Ligand Preparation:

Using Abl crystal structure the Ligprep module of Schrödinger v9.2 was used. LigPrep provides hydrogen atoms as well as creates tautomers, ionization states, ring conformers and stereoisomers, thereafter, minimizes such structures with assistance of OPLS2005 force field. Therefore, the titration with fractions from pH value of the body fluids (pH 7.0 $\pm$ 2.0) was fixed by the means of EPIK module. For the purpose of transmitting these compounds, this medium was desalinated and this allowed no tautomers to be generated. The matching chiralities were also preserved by using the same approach as for each ligand. In addition, each subunit's low vibrational energy conformation was found. The VS criteria were applied to the screened compounds while using Schrödinger's LigFilter module in version 9.2. Compounds with nomenclature such as molecular weight less than or equal to 400, number of H-bond donors less than or equal to 3, number of H-bond acceptors less than or equal to 7, number of rotatable bonds less than or equal to 8, and LogP less than or equal to 5 are considered challenging in VS [28].

Ligand Docking:

Precision of GLIDE does consider 3 types of docking accuracy. The very first step conducted was high-throughput virtual screening (HTVS) and then came the standard-precision (SP) docking which was followed by extra-precision (XP) docking to determine the best prospects⁵⁰. HTVS is used for high-throughput screening of compound libraries, which is necessary when many chemicals have to be checked rapidly. Moreover, the post docking analysis of HTVS docking was done to export and determine the scores of docking poses with docking scores (or free energy) as well as SP docking was done. Ligand was selected as XP docking input which had the highest docking scores above the cutoff level. XP docking will result in higher matching VS with no false-positive cases seen through improved docking simulations [19].

ADME/T Analysis:

The computed Qikprop 2.3 included in the Schrödinger software to determine the submitted ADME/T properties of the evaluated compounds. The Qikprop parameterizing determines fragments that are active entities in the drug molecules findings. Hence, the calculations go beyond the estimation of only the most common 95%. It allows us to comprehend the take-on mechanism of ligands by getting a hold of their molecular features which extensively describe permeability through MDCK cells, and minimal-duration antiparasitic drugs shown

by QikProp log values for channel blockage of K⁺, and QikProp violations of Lipinski's rule of five. ^[18]

ADME/T study of the drugs' candidates determines those drugs with the best pharmacokinetics, minimum toxicity, and strong efficacy. This evaluation is so significant during the searching for initial test-bed compounds with probably strong potential to be optimized than their currently low potency for effective cancer treatment. The application of ADME/T prediction to research is one of the best way scientists can select the molecule with the potential of early success. Thus, it will help to improve the efficiency of preclinical and clinical studies and also gives an opportunity to discover new powerful anti-cancer kinase BCR-ABL inhibitors. All hit compounds were evaluated using Lipinski's rules and absorption, distribution, metabolism, excretion, and toxicity (ADMET) based virtual screening. ^[29]